

Prabhat Tripathi

Full-Stack Developer | Competitive Programmer | Agentic AI Learner

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EDUCATION

K.R. Mangalam University, Gurugram

B.Tech in Computer Science Engineering

2023 – 2027

CGPA: **7.6**

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript

Backend: Node.js, Express.js, REST APIs, JSON

Frontend: React.js, HTML, CSS, Tailwind CSS

Databases: MySQL, SQL Server (stored procs, joins, triggers, indexing) PostgreSQL, MongoDB

AI / ML: Prompt Engineering, RAG, MCP, Tool Calling, LangChain, Agentic AI, Fine-Tuning

Tools: Git, GitHub, VS Code, Postman, AutoCAD

Core CS: DSA, OOPs, DBMS, OS, Computer Networks

DSA & COMPETITIVE PROGRAMMING

- Solved **300+** problems on LeetCode & Codeforces covering Dynamic Programming, Graphs, Trees, Sliding Window, and Binary Search.
- Strong command over **time & space complexity** analysis; consistently optimise solutions from brute-force to optimal.
- Actively participate in **inter-college competitive programming** contests, technical fests, and hackathons (2023–Present).
- Mentor peers on core DSA topics – arrays, recursion, backtracking, heaps, and tries – via collaborative sessions (2024–Present).

PROJECTS

NexusAssess – AI Interview Prep

Feb 2026

React · Node.js · Express.js · REST APIs

🌐 prabhat-11/NexusAssess

- Built a scalable full-stack AI platform for structured technical interview preparation with evaluation modules.
- Designed backend REST APIs and integrated AI-driven assessment generation using prompt engineering techniques.
- Deployed on Render & Vercel with CI/CD via GitHub.

Shop Monitoring System

2024

JavaScript · MySQL · ASP.NET Core · REST APIs

🌐 prabhat-11/Shop-Monitoring

- Built a responsive web app automating inventory and sales management with real-time CRUD operations.
- Integrated MySQL/SQL Server with stored procedures, indexing, and triggers for structured data tracking.
- Implemented RBAC, session management & authentication for secure enterprise multi-user access control.

OBJECTIVE

Motivated CSE student with strong foundations in **Data Structures & Algorithms**, full-stack development (React, ASP.NET Core, Node.js), and databases (SQL Server, MySQL, MongoDB). Experienced building scalable REST APIs with C# and Entity Framework. Actively exploring **Agentic AI**, RAG pipelines, and LLM fine-tuning. Passionate about writing clean, optimised, production-ready code that solves real-world problems.

CURRENTLY LEARNING – AGENTIC AI

- LLM Internals** & Prompt Engineering (chain-of-thought, few-shot, system prompts)
- Tool Calling** & Function-Use Patterns for building autonomous AI Agents
- Agentic AI Systems – planning loops, memory management, and self-reflection
- Model Context Protocol (MCP)** for structured AI-tool communication and integration
- RAG** – Retrieval-Augmented Generation with vector databases (Pinecone, ChromaDB)
- AI Frameworks: LangChain and LlamaIndex for building production-grade AI pipelines
- Working with **Vision & Multimodal** Models for image understanding and generation tasks
- Fine-Tuning** LLMs using LoRA and PEFT strategies for domain-specific adaptation

CERTIFICATIONS

- DSA – Professional Certification
- Python Essentials – Cisco Oct 2024
- C++ Programming Certification Jun 2024

KEY COURSES

DSA · OOPs · DBMS · OS · Computer Networks · Software Engineering · System Design · Discrete Mathematics · Digital Electronics · Numerical Methods · Computer Graphics & Multimedia

POSITIONS OF RESPONSIBILITY

- Mentor peers in DSA, full-stack web dev & .NET Core concepts through weekly collaborative coding sessions (2024–Present)
- Represent department in competitive programming contests, technical fests & inter-college hackathons (2023–Present)

MISCELLANEOUS

- Open-source contributor; exploring CLI utilities, developer tools & productivity automation (2023–Present)
- Continuously upskilling through online courses, tech blogs, and hands-on backend & system-design projects
- Enthusiastic about Artificial Intelligence, cloud infrastructure & distributed systems (2024–Present)